

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 September 2026
Team ID	SWTID-2026-4440
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Auto Ticket Classification using Flow Designer

Reference: <https://www.servicenow.com/docs/bundle/zurich-application-development/page/build/flow-designer/concept/flow-designer.html>

Auto Ticket Classification using Flow Designer

Technical architecture · ServiceNow Now Platform · Application scope: Global · Deployed through Update Sets

1 Presentation Layer — channels and user interfaces

Agent Workspace

The Service Desk agent reviews, confirms or overrides the predicted category and assignment group.

[Now Platform UI](#)

Service Portal

End users raise a ticket and track it; every submitted ticket enters the same classification flow.

[Service Portal / ESC](#)

Inbound Email & Virtual Agent

Tickets arriving by email or chat become Incident records that the flow trigger picks up.

[Email / Virtual Agent](#)

2 Application Logic — Flow Designer (Process Automation)

TRIGGER

Record Created — incident

LOOK UP

Classification rule table

PREDICT

Category & assignment group

DECIDE

Confidence $\geq$ 85%

ACT

Auto-route or triage

Flow: Auto Ticket Classification

Built on the Flow Designer canvas as a flow with three subflows (notify, audit, fallback) and a custom action that scores keywords against the rule table. All of it is captured in the current Update Set.

[Flow Designer](#) · [Subflows](#) · [Custom Actions](#) · [Flow API](#)

3 Data Layer — ServiceNow tables and records

incident

Out-of-the-box ticket table. Category, subcategory and assignment group are written back here.

[OOB table](#)

Incident WorkFlow

Custom table created for this project that stores the incoming ticket data handled by the flow.

[u_incident_work_flow](#)

Classification rules

Keyword-to-category mapping used by the look-up step and the keyword scorer.

[u_ticket_classification_rule](#)

Audit log & groups

Every prediction and outcome is logged; assignment groups drive the routing decision.

[u_classification_log](#) · [sys_user_group](#)

4 Platform & Integration Services

Predictive Intelligence

Classification solution trained on past closed incidents; predicts category and assignment group.

[ServiceNow ML](#)

IntegrationHub & REST API

Inbound REST / Table API lets external monitoring tools open incidents automatically.

[REST](#) · [Webhook](#) · [Import Set](#)

Notification service

Emails the assignment group after routing and raises alerts when confidence is low.

[Email Notification](#)

Update Set

Version control for the flow, subflows, actions and the table dictionary.

[Project Update Set](#)

5 Infrastructure — hosting and access

Now Platform cloud

ServiceNow-hosted, multi-tenant cloud. No local server is required to run the classification flow.

[ServiceNow Cloud](#)

Instances & scopes

Sub-production instance for build and test, then promoted to production. Everything sits in the Global scope.

[Dev](#) · [Test](#) · [Prod](#)

Access control

Role-based ACLs and SSO restrict who may read the rule table and override a prediction.

[ACL](#) · [SSO](#) · [Roles](#)

Figure 1. Technical architecture of the Auto Ticket Classification solution — five layers from the agent-facing interfaces down to the ServiceNow cloud infrastructure, with Flow Designer at the centre of the application logic.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the application — agent workspace, self-service portal and virtual agent	ServiceNow Agent Workspace; Service Portal; Virtual Agent (Now Platform UI)

S.No	Component	Description	Technology
2	Application Logic-1	Logic that classifies an incoming ticket and routes it to the correct group	ServiceNow Flow Designer — flow “Auto Ticket Classification”
3	Application Logic-2	Reusable automation steps shared across flows	Flow Designer Subflows (notify, audit, manual-triage fallback)
4	Application Logic-3	Custom scoring logic used when the ML model cannot decide	Flow Designer Custom Action + Script step (JavaScript, GlideRecord)
5	Database	Data type and configuration — the tables the flow reads and writes	ServiceNow tables: incident (OOB), u_incident_work_flow (custom), u_ticket_classification_rule, u_classification_log
6	Cloud Database	Database service on cloud	ServiceNow Cloud — Now Platform instance database (no external DB required)
7	File Storage	File storage requirements — attachments added to a ticket	ServiceNow Attachment table (sys_attachment), stored on the Now Platform cloud
8	External API-1	Purpose of the external API used to create tickets and request a prediction	ServiceNow REST / Table API (inbound); Predictive Intelligence API (sn_ml)
9	External API-2	Purpose of the external API used for notification and collaboration	IntegrationHub REST step — SMTP email and Microsoft Teams notification
10	Machine Learning Model	Purpose of the Machine Learning model — predict Category, Subcategory and Assignment group, with a confidence score	ServiceNow Predictive Intelligence — classification solution trained on historical closed incidents
11	Infrastructure (Server / Cloud)	Application deployment on local system / cloud Local Server Configuration: none — the platform is SaaS only Cloud Server Configuration: ServiceNow Now Platform, multi-tenant cloud, Global application scope, Dev → Test → Prod instances	ServiceNow Cloud; Update Sets for promotion between instances

Table-2: Application Characteristics:

S.No	Component	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used	None required — the solution is built entirely on the ServiceNow platform. JavaScript (Rhino engine) inside Flow Designer Script steps; REST / JSON for integration. The Now Platform UI Framework can be used if a custom React widget is added.

S.No	Component	Description	Technology
2	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	Role-based Access Control and table-level ACLs on u_incident_work_flow and the rule table; SSO / SAML and multi-factor sign-on; TLS 1.2+ in transit and encryption at rest on the platform; audit logging of every classification and override; IP allow-list on the inbound REST endpoint.
3	Scalable Architecture	Justify the scalability of architecture (3-tier, micro-services)	Tiered ServiceNow architecture (presentation / logic / data) delivered as a multi-tenant SaaS service. Flow Designer executes asynchronously off the insert transaction with queued and parallel execution, so throughput scales with the instance rather than with any custom server.
4	Availability	Justify the availability of the application (e.g. load balancers, distributed servers)	ServiceNow cloud SLA with redundant data centres and automatic failover across availability zones. Flow steps retry on failure, and any low-confidence ticket falls back to the manual triage queue, so classification never blocks ticket creation.
5	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDNs)	The flow trigger is asynchronous, so the ticket insert is not delayed; key fields on the rule table are indexed for fast look-ups; predictions are cached per session. Target: classification completed in under 2 seconds per ticket at approximately 50 tickets per minute per instance.

ServiceNow Configuration – Create the Custom Table

The custom table that holds the ticket data classified by the flow is created on the ServiceNow platform before the flow itself is built. The table is named Incident WorkFlow (u_incident_work_flow) and it is created inside the Global application scope, so every dictionary change is captured in the current update set (Project Update Set).

Steps to create the table:

1. Open **All** menus → Search for “Tables”
2. Under **System Definition** → Click **Tables**
3. On the top right corner click the **New** button
4. **Label:** Incident WorkFlow
5. **Uncheck** the “Create module” checkbox
6. Now **Save** the Form
7. Then click the **Controls** section and enable the checkbox “**Auto Number**” to **true**
8. Now again **save** the Form
9. Then Save the Form and under **Related Links** click “**Show Form**”
10. Now open the **Configure** option on clicking the **Context Menu** and then open the **Form Designer**
11. Then **create fields** as required (Number, Short description, Category, Subcategory, Assignment group, Confidence, Auto classified and Classification notes)

The screens below follow the same red-callout convention as the reference images: the numbered red tag marks the exact control used at each step.

1

All

servicenow.

Favorites

History

Workspaces

Admin

Application scope: Global

Update set: Project Update Set

AB

All > Application Navigator > System Definition

Search the Application Navigator for Tables

Type **Tables** in the navigator filter → under **System Definition** click **Tables**, then click **New**.

2

Q Tables |

System Definition > Dictionary

System Definition > **Tables** [sys_db_object]

System Definition > Table Columns

System Definition > Choice Lists

Tables

3New

Label String	Name String	Extends table String	Application String
Task	task	—	Global
Incident	incident	task	Global
Problem	problem	task	Global
Change Request	change_request	task	Global
Classification Rule	u_ticket_classification_rule	—	Global

1

Open **All** menus (top left hamburger) to reach the Application Navigator.

2

Search for **Tables** and, under **System Definition**, click **Tables**.

3

On the **Tables** list, click **New** in the top right corner of the page.

Figure 2. Steps 1–3: locating **System Definition > Tables** in the Application Navigator and opening the new-record form with the **New** button.

Figure 2. Steps 1–3: opening All menus, searching for Tables in the Application Navigator, and clicking New on the Tables list.

servicenow.

All

Favorites

History

Workspaces

Admin

Application scope: Global

Update set: Project Update Set

AB

System Definition > Tables > New record

Table

Label

Incident WorkFlow

Name

u_incident_work_flow (derived from the label)

Extends table

— None —

Create module

☐ false (unchanged)

Create menu

☐ false

Add module to menu

— None —

Application

Global

Access controls

Maintain the default ACLs generated for the table

6 Save

Saving creates the table, its sys_db_object entry and the dictionary records.

4 Set the Label to Incident WorkFlow.

5 Uncheck the Create module checkbox.

6 Now Save the form.

Figure 3. Steps 4–6: the new Table record for Incident WorkFlow. Create module is left unchecked, and the form is saved to create the table.

Figure 3. Steps 4–6: naming the table Incident WorkFlow, leaving Create module unchecked, and saving the form.

servicenow.

All

Favorites

History

Workspaces

Admin

Application scope: Global

Update set: Project Update Set

AB

System Definition > Tables > Incident WorkFlow

Incident WorkFlow

u_incident_work_flow · Application: Global · saved

Table

Controls

Application Access

Related Links

Controls

Auto Number

☒ true

each new record gets a unique number

Live feed

☒ true

Audit

☐ false

Accessible from

All application scopes

Extends

— None —

Text indexing

Index all fields of the table

8 Save

Enabling Auto Number adds the Number field and its numbering counter to the table.

7 Open the Controls section and enable the Auto Number checkbox — set it to true.

8 Now save the form again, so the numbering record is created for the table.

Figure 4. Steps 7–8: the Controls section of the Incident WorkFlow table with Auto Number set to true, then saved.

Figure 4. Steps 7–8: enabling Auto Number to true in the Controls section, then saving the form again.

9

Incident WorkFlow INCWF0001001

System Definition > Tables > Incident WorkFlow

Incident WorkFlow	
Label	Incident WorkFlow
Name	u_incident_work_flow
Number	INCWF0001001
Auto Number	☒ true
Application	Global

10

Show Form

Show List

Add to Update Set

Personalize Dictionary

CONTEXT MENU — RIGHT-CLICK THE FORM HEADER

CONFIGURE — SUBMENU

11

Save

Insert

Insert and Stay

Analyze Access

Configure

Export

View

Create Favorite

Copy URL

Copy sys_id

Show XML

History

Reload form

12

Form Builder

Form Design

Form Layout

Related Lists

All

Table

Security Rules

Business Rules

Client Scripts

UI Policies

Data Policies

UI Actions

Notifications

Dictionary

Update

Delete

9

Save the form, then use the **context menu** (hamburger / right-click) on the record.

10

Under **Related Links**, click **Show Form** to open the table form for the new table.

11

Open the **Configure** option from the context menu.

12

Then open the **Form Design** (Form Designer) option.

Figure 5. Steps 9–12: opening the **Incident WorkFlow** form through **Related Links > Show Form** and reaching **Configure > Form Design** from the context menu — the same red-callout style as the reference screenshots.

Figure 5. Steps 9–10: opening the form through Related Links → Show Form and reaching Configure → Form Design from the context menu.

servicenow. All Favorites History Workspaces Admin
Application scope: Global Update set: Project Update Set AB

System Definition > Tables > Incident WorkFlow > Form Design

Form Designer — u_incident_work_flow

Drag the field types from the palette and drop them on the form to build the layout that the flow reads and writes.

Field Types

- ☐ String
- ☐ Choice
- ☐ Reference
- ☐ Integer
- ☐ True/False
- ☐ Date/Time
- ☐ Journal

Incident WorkFlow · form layout

Number	String	auto-numbered
Short description	String	the ticket text
Category	Choice	predicted by the flow
Subcategory	Choice	predicted by the flow
Assignment group	Reference	sys_user_group
Confidence	Integer	0 – 100 %
Auto classified	True/False	true when ≥ 85 %
Classification notes	Journal	audit trail

Field Navigator

- Number
- Short description
- Category
- Subcategory
- Assignment group
- Confidence
- Auto classified
- Classification notes

13 Create the fields required by the table in Form Designer and arrange them on the form layout.

Figure 6. The Form Designer layout for `u_incident_work_flow` — the fields the Flow Designer flow writes to once a ticket has been classified.

Figure 6. Step 11: the Form Designer layout for `u_incident_work_flow`, with the fields the classification flow reads and writes.

Once the table and its fields exist, the Flow Designer flow is built against them: the flow triggers on record creation, looks up the classification rules, predicts the category and assignment group, and writes the result back to Category, Subcategory, Assignment group and Confidence on this table.

References:

- <https://www.servicenow.com/docs/bundle/zurich-application-development/page/build/flow-designer/concept/flow-designer.html>
- https://www.servicenow.com/docs/bundle/zurich-platform-administration/page/administer/table-administration/concept/c_CreatingCustomTables.html
- <https://www.servicenow.com/docs/>
- <https://developer.servicenow.com/dev.do>
- <https://c4model.com/>
- <https://aws.amazon.com/architecture>
- <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>